

AARON KARSTEN

aaronkarsten01@gmail.com | Tempe, AZ | linkedin.com/in/aaron-karsten | github.com/aaronk2001
Full-Stack Robotics Engineer | Fleet-Scale Production Automation | PLC • PCB • Python • Computer Vision

SUMMARY

Full-stack robotics engineer (mechanical, electrical, software) who built an in-house production fleet from **0 to 200 writing robots** and scaled output **3x to 30,000 letters/day**. Designed automated inspection and manufacturing machines end to end, shipped 4 custom PCBs fleet-wide, and deployed YOLO/PyTorch vision inspection across production. Promoted from intern to Engineer II in 10 months; held the fleet at **96% uptime with a 3-person team**.

PROFESSIONAL EXPERIENCE

Handwrytten

Tempe, AZ

Designs and builds proprietary robotic handwriting machines fully in-house, writing pen-on-paper letters at scale.

Robotics Engineer II

Jul 2024 – Sep 2026

Robotics Engineer Intern (promoted within 10 months)

Sep 2023 – Jun 2024

- Scaled production capacity **3x — 10,000 to 30,000 letters/day** — by designing and developing highly automated manufacturing and QA machines from concept to production, growing the in-house fleet from **0 to 200 writing robots**, plus 6 QA robots and 3 production-support robots.
- Sustained **96% uptime** across the 200-robot fleet with a **3-person team** (down from the 6-person ramp crew) by running preventive maintenance, covering on-call incident response, and engineering reliability into the fleet.
- Led a **6-person** cross-functional ramp team (2 engineers, 4 technicians) building **6 machines/week** at peak, owning machine specifications, manufacturing instructions, and on-time delivery through supply-chain volatility.
- Cut labor cost **\$100K+/year** and removed the manual-inspection bottleneck behind the 3x scale-up by designing and building **2** automated machines on Arduino Opta and Portenta Machine Control PLCs that inspect, sort, and print letters and notes — defining process flow and inspection methodology end to end against a **3-5% defect rate**.
- Eliminated all per-job operator setup across **200 robots** by co-authoring the multi-threaded Python control application on every machine — fully autonomous operation from one barcode scan with dynamic job pulling.
- Cut mean-time-to-repair **65% (10 → 3.5 hours)** by building real-time telemetry, fault detection, and diagnostics with Prometheus and Grafana across the production fleet.
- Shipped **4 custom PCBs** to fleet-wide deployment — a Raspberry Pi HAT robot controller, a combined power distribution and logic board, a stepper motor driver board, and a Pico relay board — owning the full cycle in EasyEDA from schematic capture through fabrication.
- Co-developed a fleet-wide **OEE tracking system**: every machine posts live status to an AWS/SQL pipeline computing availability, performance, and quality — sustaining **98.7% average quality** fleet-wide.
- Deployed **YOLO/PyTorch** vision models on fleet Linux embedded systems, inspecting **30,000 letters/day** in real time for ink skips, ink color, smudges, text misalignment, and wrong card or envelope.
- Cut custom-part lead time from **4+ weeks to under 1 week** by standing up an in-house fabrication cell (FDM 3D printing, laser cutting) producing **500+ parts/month**, helping hold unit cost flat through tariff increases.

PROJECTS

- **Los Alamos National Laboratory (via ASU) — Project Manager**: Delivered a 6-DOF UR5e robotic-arm-actuated glovebox for hazardous-material handling, hitting **100% of milestones over an 8-month project** (Aug 2023 – Apr 2024) leading a 3-person interdisciplinary team — RoboDK simulation, SolidWorks prototyping, and dual-mode control (URScript + flight-stick digital twin via a custom Python bridge).
- **Modular Robot Arm (personal, in progress)**: Building a robot arm full stack, solo — modeling the mechanical design in CAD, designing a custom KiCAD joint controller PCB that combines BLDC motor control with a CAN FD bus (TCAN1462-Q1 transceiver), and programming the control software.

TECHNICAL SKILLS

Robotics & Controls: PLC programming (Codesys, Arduino Opta, Portenta Machine Control, ladder logic), Factory I/O, UR5e / PolyScope, URScript, RoboDK, motion control, HMI/SCADA, OEE & fleet telemetry

Software & ML: Python (multi-threaded control systems), PyTorch, YOLO, computer vision, C/C++, Bash, MATLAB, SQL, Linux, Git, Docker, AWS, Prometheus, Grafana, Jira

Electrical: EasyEDA, KiCAD, PCB design & fabrication, CAN FD, BLDC motor control, Cadence, control wiring, network cabling, soldering

Mechanical: SolidWorks, Fusion 360, OpenSCAD, FDM 3D printing, laser cutting, DFM/DFA from production builds

EDUCATION

B.S.E., Robotics Engineering — Arizona State University, Ira A. Fulton Schools of Engineering

December 2024